

# 1 Supplemental Information

## 2 1. VISCOELASTIC HYPOTHESES

3 Here, we apply simple models of material properties to predict the physi-  
4 cal impact of sporulation on biofilms. Each model addresses a hypothesis  
5 for the effect of sporulation. The first hypothesis, H1, is that spores act  
6 as rigid inclusions (i.e., particles interspersed throughout a polymer net-  
7 work that bolster its stiffness). We address H1 with the rule of mixtures  
8 (see Section 1A). The second hypothesis, H2, is that sporulation reduces  
9 the number of vegetative cells which in turn reduces the quantity and  
10 density of extracellular matrix. We use gelation theory to address H2  
11 (see Section 1B). The third hypothesis, H3, is that sporulation causes  
12 the polymer network to restructure more easily (i.e., for the processes  
13 influencing cross-links, entanglements, and other network parameters to  
14 have a more active role). We address H3 using soft glassy rheology.

### 15 A. Rule of mixtures

16 In order to make predictions based on H1 (that spores act as rigid inclu-  
17 sions) we utilize the rule of mixtures. The rule of mixtures is used in  
18 materials science to estimate upper- and lower-bounds for the physical  
19 properties, including stiffness, of composite materials. Here, we treat  
20 vegetative cells and extracellular matrix as one material and spores as  
21 another, making the biofilm a composite material consisting of cells + ma-  
22 trix and spores. We group vegetative cells and matrix in a distinct group  
23 from spores due to the similarity in their independent elastic moduli and  
24 because their moduli are many orders of magnitude smaller than that of  
25 spores (1, 2).

$$G^* = fG_1^* + (1 - f)G_2^* \quad (S1)$$

$$G^* = \left( \frac{f}{G_1^*} + \frac{1 - f}{G_2^*} \right)^{-1} \quad (S2)$$

26 For a composite material composed of two constituents, the composite  
27 complex modulus  $G^*$  is predicted by Eqs. S1 and S2, where  $G_1^*$  and  $G_2^*$   
28 are the complex moduli of the constituents and  $f$  is the volume fraction  
29 of the first constituent. Equation S1, the rule of mixtures, is derived from  
30 the Voigt model and assumes that the strain is equal in both constituents.  
31 This equation sets the upper bound on the composite modulus. Equation  
32 S2, the inverse rule of mixtures, is derived from the Reuss model and  
33 assumes that the stress is equal in both constituents. This equation sets

the lower bound on the composite modulus. Here, we note that real materials must fall somewhere between the Voigt and Reuss models (i.e., somewhere between having constituent materials with equal strain or stress) although few materials of interest are likely purely Voigt- or Reuss-like. This allows for upper and lower bounds to be placed on composite material properties.

Both Eqs. S1 and S2 predict an increase in  $G^*$  with an increase in  $f$ . Additionally, both predict increases in  $G'$  and  $G''$ , the storage (elastic) and loss (viscous) moduli, with an increase in  $f$ . In the case of bacterial biofilms,  $f$  would represent the volume fraction of spores and  $1 - f$  the volume fraction of everything else (i.e., the vegetative cells and extracellular matrix). This would mean that as the spore fraction increases, so too would the composite modulus, with greater spore  $G'$  and  $G''$  leading to a more pronounced effect. Spores are known to have a greater elastic modulus than bulk matrix (approximately  $10^6 \times$ ) (1, 2), so spore fraction would have a large effect on the composite  $G'$ . In contrast, spores do not have a well-defined independent viscosity, making their contribution to the composite  $G''$  minimal. For these reasons, H1 predicts that knocking out sporulation ( $\Delta P_s$ ) would decrease stiffness. However, our data show the opposite trend, with moduli decreasing in spore-producing biofilms, indicating that H1 is insufficient to explain the role of sporulation in biofilm material properties.

#### A.1. Derivations

Here we derive Eqs. S1 and S2.

$$\gamma = \gamma_1 = \gamma_2 \quad (\text{S3})$$

$$\sigma = G^* \gamma \quad (\text{S4})$$

In the Voigt model, we assume the constituents have equal strain (Eq. S3) and apply Hooke's law (Eq. S4).

$$\sigma = f\sigma_1 + (1 - f)\sigma_2 \quad (\text{S5})$$

$$G^* \gamma = fG_1^* \gamma_1 + (1 - f)G_2^* \gamma_2 \quad (\text{S6})$$

$$G^* = fG_1^* + (1 - f)G_2^* \quad (\text{S7})$$

With forces balanced at the interface between materials, we arrive at Eq. S5, which when combined with Eqs. S3 and S4 yields Eq. S7, which is identical to Eq. S1.

$$\sigma = \sigma_1 = \sigma_2 \quad (\text{S8})$$

In the Reuss model, we assume the constituents have equal stress (Eq. S8) and utilize Hooke's law (Eq. S4).

$$\gamma = f\gamma_1 + (1 - f)\gamma_2 \quad (\text{S9})$$

$$\frac{\sigma}{G^*} = \frac{f\sigma_1}{G_1^*} + \frac{(1 - f)\sigma_2}{G_2^*} \quad (\text{S10})$$

$$\frac{1}{G^*} = \frac{f}{G_1^*} + \frac{1 - f}{G_2^*} \quad (\text{S11})$$

$$G^* = \left( \frac{f}{G_1^*} + \frac{1 - f}{G_2^*} \right)^{-1} \quad (\text{S12})$$

65 Strain distributes between the constituents according to Eq. S9, which  
 66 when combined with Eqs. S8 and S4 yields Eq. S12, which is identical to  
 67 Eq. S2.

68 See *Polymer Science Dictionary* by Alger (3) and Abdalla (4) for addi-  
 69 tional information.

## 70 B. Networks in gelation regime

71 Hypothesis 2 predicts that sporulation's primary effect on biofilm rheo-  
 72 logical properties is to reduce the density of the colony's matrix network:  
 73 if more cells become dormant spores, fewer cells are in a state a matrix  
 74 production, resulting in a less dense polymer network. To predict what  
 75 effect this would have on rheological measurements, we must relate net-  
 76 work density to measured moduli, a pursuit to which percolation theory  
 77 lends itself well. This approach is defensible since *Bacillus subtilis* biofilms  
 78 are known to be near the sol-gel percolation point (5).

$$G \approx \frac{\rho RT}{M_x} + T_e G_e \quad (\text{S13})$$

79 Percolation theory predicts that, for a polymer network in the gelation  
 80 regime, elastic modulus scales with network density, as described by Eq.  
 81 S13 (6), where  $G$  is the elastic modulus,  $\rho$  the network density,  $R$  the gas  
 82 constant,  $T$  the temperature,  $M_x$  the number-average apparent molar  
 83 mass of a network strand,  $T_e$  the entanglement trapping factor, and  $G_e$   
 84 the entanglement modulus.

85 Equation S13 predicts that elastic modulus  $G$  decreases with a decrease  
 86 in network density  $\rho$ . In the case of biofilms, having more spores may  
 87 correlate with having a lower density of extracellular matrix (compared  
 88 to comparable biofilms without spores). Therefore, according to Eq. S13,  
 89 spore-producing biofilms would have a lower elastic modulus.

$$\epsilon = \frac{\rho - \rho_c}{\rho_c} \quad (\text{S14})$$

$$G \sim \epsilon^t \quad (\text{S15})$$

90 Assuming that biofilm polymer networks are not far from the perco-  
 91 lation point (as demonstrated in (5)), it is convenient use the parameter  
 92  $\epsilon$  as defined in Eq. S14 (7) to write an expression for  $G$  as in Eq. S15 (7).  
 93 The parameter  $\rho$  here represents the network density and  $\rho_c$  the gelation  
 94 critical density. The scaling exponent  $t$  (not time) dictates how  $G$  changes  
 95 with respect to  $\epsilon$ .

$$\eta \sim \epsilon^{-s} \quad (\text{S16})$$

96 Percolation theory allows us to write a similar expression for the dy-  
 97 namic viscosity  $\eta$ : Eq. S16 (7). Here  $s$  is a scaling exponent representing  
 98 how strongly network density lowers viscosity.

99 From Eqs. S14 - S16 we see that a decrease in network density  $\rho$   
 100 corresponds to a decrease in  $G$  and an increase in  $\eta$ .

$$H(\tau) \sim \tau^{-n} \text{ for } \tau \ll \tau_z \quad (\text{S17})$$

101 In the gelation regime, percolation theory also predicts that the relax-  
 102 ation spectrum follows Eq. S17 (8). Here,  $H(\tau)$  represents the relaxation  
 103 spectrum,  $\tau$  is the relaxation time,  $n$  is a scaling exponent, and  $\tau_z$  is the  
 104 relaxation spectrum cutoff time. Above the gelation point, the theory  
 105 predicts a stable long-term elastic contribution as  $\tau$  approaches infinity:  
 106  $G_{inf}$ .

$$\tau_z \sim |\epsilon|^{-\Delta} \quad (\text{S18})$$

107 The cutoff relaxation time  $\tau_z$  is defined by Eq. S18 (9) where  $\epsilon$  is the  
 108 distance from the critical gel point density  $\rho_c$  and  $\Delta$  is a class scaling  
 109 exponent. The quantity  $\epsilon$  varies with network density according to Eq.  
 110 S14.

111 From Eqs. S17 - S14 we see that a decrease in network density  $\rho$ , as  
 112 from sporulation, results in an increase in the cutoff relaxation time  $\tau_z$   
 113 leading to a broader relaxation spectrum  $H(\tau)$ . Additionally, a looser  
 114 relaxation spectrum may be interpreted as a more disordered network.

### 115 C. Soft glassy rheology

116 Soft glassy rheology (SGR) is a phenomenological model of material  
 117 properties (10, 11). This model consists of mesoscopic elements that  
 118 transition between energy "traps" in response to strain.

$$E' = E - \frac{1}{2}kl^2 \quad (\text{S19})$$

$$\rho(E') \sim e^{-E'} \quad (\text{S20})$$

$$P_{eq}(E') \sim \rho(E')e^{-E'/x} \quad (\text{S21})$$

119 The depth of the traps (defined by Eq. S19) determines the strain  $l$   
 120 necessary for an element to escape the trap and fall into one of different  
 121 energy. The energy landscape of these traps is denoted by  $\rho(E')$  and takes  
 122 the form of exponential decay as shown in Eq. S20. In the soft glassy  
 123 fluid regime, which is the phase most applicable to biofilms (12–14),  
 124 the material's probability distribution evolves towards the Boltzmann  
 125 distribution as defined by Eq. S21.

126 The parameter  $x$ , known as the "noise temperature", is a mean-field  
 127 subsumption of the processes through which elements transition between  
 128 traps. The transition of elements between traps is what we refer to as  
 129 network restructuring, although in reality it likely includes this and other  
 130 processes. The noise temperature is key in determining if the material  
 131 behaves like a true glass ( $x < 1$ ), a soft glassy fluid ( $1 < x < 2$ ), an  
 132 ordinary Newtonian/Maxwell-like fluid ( $x = 2$ ), or a simple liquid  
 133 ( $x > 2$ ).

$$G'(\omega) \sim G'_0 \left( \frac{\omega}{\omega_0} \right)^{x-1} \quad (\text{S22})$$

$$G''(\omega) \sim G''_0 \left( \frac{\omega}{\omega_0} \right)^{x-1} \quad (\text{S23})$$

134 Under SGR (in the soft glassy regime:  $1 < x < 2$ ), the shear moduli  
 135 take the form of Eqs. S22 and S23.

$$P_{eq}(\tau) \sim \tau^{-x} \quad (\text{S24})$$

136 Additionally, the relaxation time spectrum exhibits power law behavior  
 137 as described by Eq. S24 for  $\tau \geq 0.1$  s.

138 We propose that sporulation allows easier element rearrangement (i.e.,  
 139 network restructuring, etc.), increasing  $x$ . With an increase in  $x$ , Eqs.  
 140 S22 and S23 predict a decrease in  $G'$  and  $G''$ , a trend qualitatively in  
 141 agreement with our data. Additionally, an increase in  $x$  results in faster  
 142 'decay' in Eq. S24, resulting in a narrower relaxation spectrum, consistent  
 143 with our observations.

$$\Gamma \sim \Gamma_0 e^{-E'/x} \quad (\text{S25})$$

144 Finally, the damage and recovery metrics laid out in the main text  
 145 can also be explained through SGR. SGR describes damage as elements  
 146 having sufficient strain to push them into different energy traps, resulting  
 147 in yielding at a rate defined by Eq. S25.

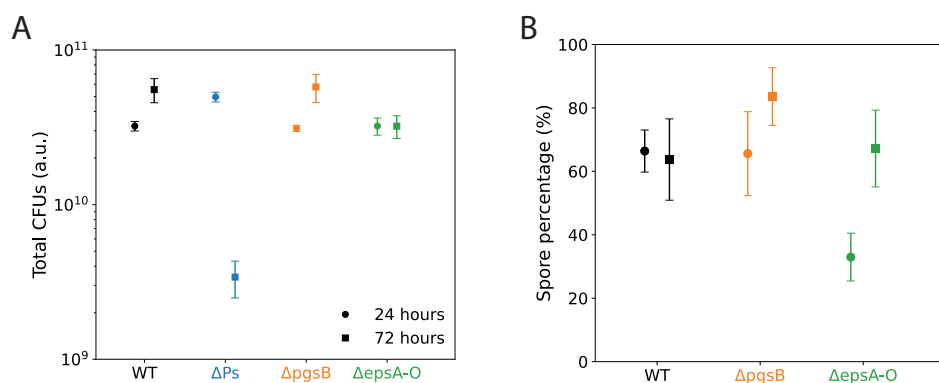
148 As  $x$  increases, as we propose in the case of sporulation, Eq. S25  
 149 predicts an increase in the element yield rate, corresponding to an increase  
 150 in damage (as defined in the main text). Predictions for recovery are less  
 151 definitive under SGR: SGR predicts that the noise temperature  $x$  affects

the recovery rate, but it doesn't make a definitive prediction of whether it increases or decreases recovery. Applying SGR in detail to further describe biofilm properties and predict the impact of sporulation is an avenue of future work.

## 2. SUPPLEMENTAL DATA

All data and analysis code can be accessed at <http://www.biophysj.org> and [https://github.com/Larkin-Lab/ducharme\\_larkin\\_sporulation.git](https://github.com/Larkin-Lab/ducharme_larkin_sporulation.git).

### A. CFU counts and spore percentages



**Fig. S1.** (A) Total CFUs: arithmetic mean and standard deviation. At 24 hours, the spore-knockout has greater total CFUs than WT (1.66×). At 72 hours, the spore-knockout has CFUs much less than 72-hour WT (0.064×) and 24-hour spore-knockout (0.07×). Wild-type and the PGA-knockout have similar total CFUs at both 24 and 72 hours, with both showing an increase with time. The EPS-knockout is constant in total CFUs from 24 to 72 hours. (B) Spore percentages: geometric mean and standard deviation. The EPS-knockout has a lower spore percentage at 24 hours compared to WT (0.77×). At 72 hours, the PGA-knockout has a higher spore percentage than WT (0.37×). N = 3 independent samples per strain.

$$N_{\text{CFUs}} = N_{\text{counts}} * 10^{n+2} \quad (\text{S26})$$

We calculated the number of CFUs using Eq. S26, where  $n$  is the power of the dilution (6 or 7) and the factor of  $10^2$  accounts for the initial and plated volumes of 10 mL and 100  $\mu\text{L}$ , respectively.

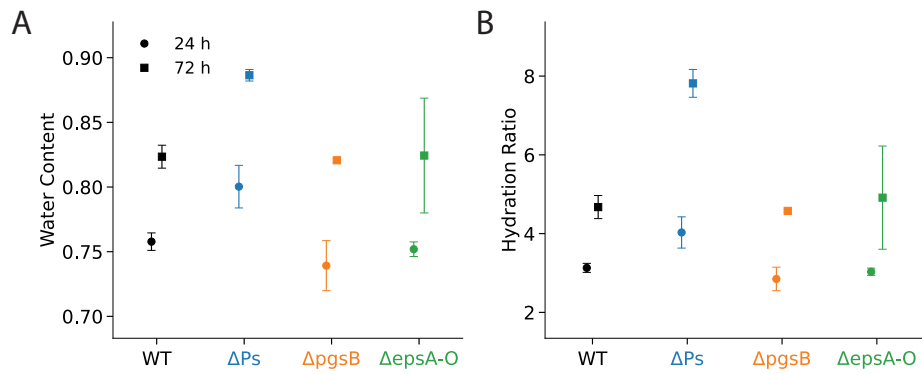
Figure S1 shows CFU counts and spore percentages for the bacterial strains used in this work. The spore-knockout has greater total CFUs than WT at 24 hours. The PGA- and EPS-knockouts have comparable total CFUs to WT at 24 hours. At 72 hours, the spore-knockout drops in

167 total CFUs substantially. The EPS-knockout has slightly lower total CFUs  
 168 than WT at that time point.

169 The WT and PGA-knockout biofilms have similar spore percentages  
 170 at 24 hours, and the EPS-knockout has a lower percentage. At 72 hours,  
 171 the PGA-knockout has a higher spore percentage than WT whereas the  
 172 EPS-knockout has a comparable percentage to WT.

173 These data indicate it is unlikely that the total CFUs or spore percentage  
 174 explain any rheological differences of the PGA- and EPS-knockouts from  
 175 WT. However, it is possible that these quantities have some indirect  
 176 impact on rheological properties. In particular the EPS-knockout, which  
 177 exhibits some differences in these quantities, may partially contribute  
 178 to the minor differences in rheological properties noted in the main text.  
 179 The spore-knockout also has substantial differences in both total CFUs  
 180 and spore percentage compared to WT which may indirectly contribute  
 181 to differences in rheological properties.

## 182 B. Mass measurements



**Fig. S2.** Plots of biofilm (A) water content and (B) hydration ratio at 24 and 72 hours. The water content is the ratio of bound water mass to wet mass. The hydration ratio is the ratio of bound water mass to dry mass. At 24 and 72 hours, spore-knockout biofilms have greater average water content and hydration ratio compared to spore-producing biofilms (WT, matrix-knockouts). All biofilms increase in both metrics from 24 to 72 hours. N = 3 independent samples per strain.

$$m_{bound} = m_{wet} - m_{dry} \quad (S27)$$

$$WC = \frac{m_{bound}}{m_{wet}} \quad (S28)$$

$$HR = \frac{m_{bound}}{m_{dry}} \quad (S29)$$

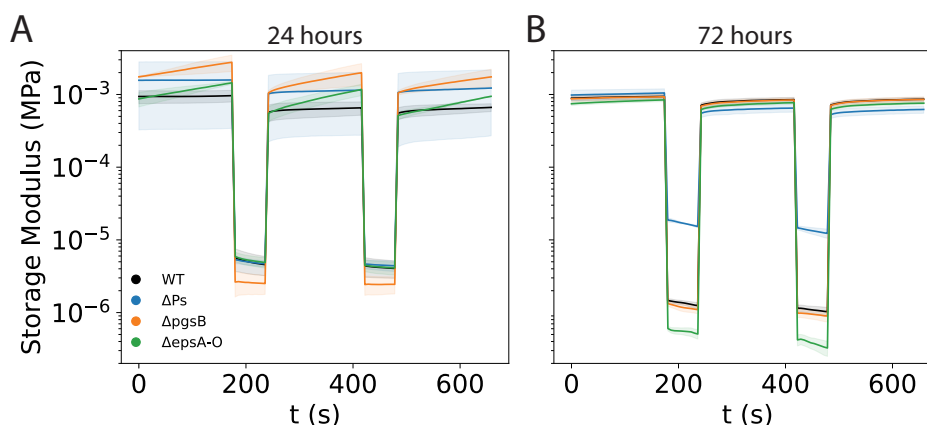
183 We calculated the mass of bound water in the biofilm as the difference  
 184 between the wet and dry masses of the biofilm (Eq. S27). We took the

185 mass of the extracellular matrix, cells, and spores as the dry mass. We  
 186 defined the water content (WC) as the ratio of the bound water mass to  
 187 the wet, or total, mass (Eq. S28) and the hydration ratio (HR) as the ratio  
 188 of bound water mass to dry mass (Eq. S29).

189 Figure S2 shows the bound water content and hydration ratio of  
 190 biofilms formed by the bacterial strains used in this work. Both metrics  
 191 are greater in the spore-knockout biofilms ( $\Delta P_s$ ) than the spore-producing  
 192 biofilms (WT,  $\Delta pgsB$ ,  $\Delta epsA-O$ ). Also, both metrics increase from 24 to 72  
 193 hours.

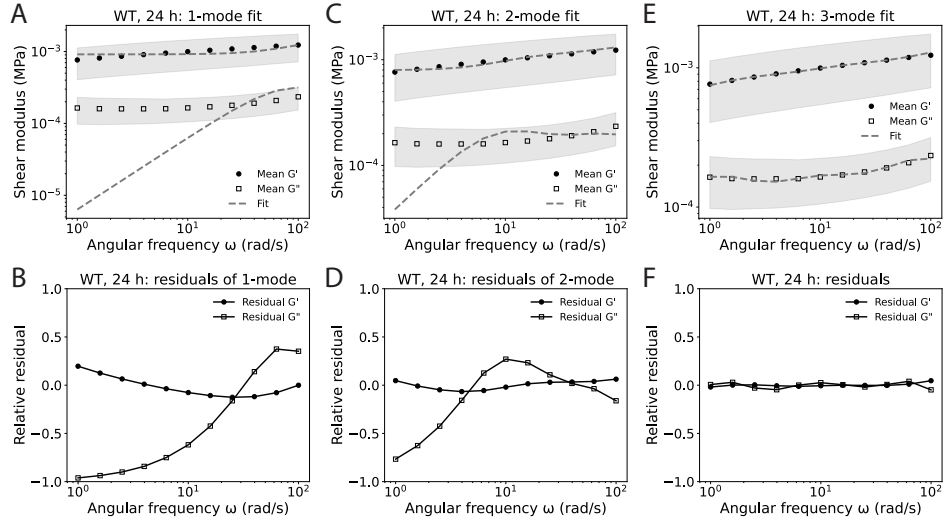
194 These data suggest bound water variations from sporulation may play  
 195 some role in biofilm rheological properties. The data show that there is  
 196 more bound water in spore-knockout biofilms. However, having more  
 197 bound water generally decreases both the storage and loss moduli of gels  
 198 and gel-like materials (15). Because we find that the storage modulus  
 199 is actually greater in the spore-knockout at 24 hours (and shows no  
 200 variation from WT at 72 hours) (see Fig. S7), bound water is insufficient  
 201 to explain our rheological data.

### 202 C. Recovery assay data



**Fig. S3.** Mean storage modulus vs time at (A) 24 and (B) 72 hours measured with alternating periods of low- and high-amplitude strain. Shaded regions indicate standard deviations. Both panels share the same y-limits.  $N = 3$  independent samples per strain.

203 Figure S3 shows the mean storage modulus vs time with standard  
 204 deviations for our four bacterial strains at 24 and 72 hours. At 24 hours,  
 205 the PGA- and EPS-knockout biofilms exhibited an upward drift during  
 206 the low-strain periods, possibly indicating that these biofilms have not  
 207 yet structural equilibrium. This data is further discussed in the main text.



**Fig. S4.** Multi-mode Maxwell fits and relative residuals. Average fit and relative residuals for one- (A, B), two- (C, D), and three-mode (E, F) generalized Maxwell models to WT storage and loss modulus data at 24 hours. The relative residuals for the one- (B) and two-mode (D) models are large, and much larger than that of the three-mode (F) model. Additionally, the lower-order fits (A, C) are qualitatively much poorer than that of the three-mode (E) model.  $N = 3$  independent samples.

#### 208 D. Frequency sweep fits

209 Figure S4 shows representative fits and relative residuals using general-  
 210 ized Maxwells models with one-, two-, and three-modes. The one-mode  
 211 model fits a mode at 0.01 s and  $G_{inf}$  and the two mode-model addition-  
 212 ally fits a mode at 0.1 s. The one- and two-mode fits are of poor quality.  
 213 The three-mode Maxwell model, which fits an additional mode at 1.0 s,  
 214 is the lowest-order model for which the fits are acceptable.

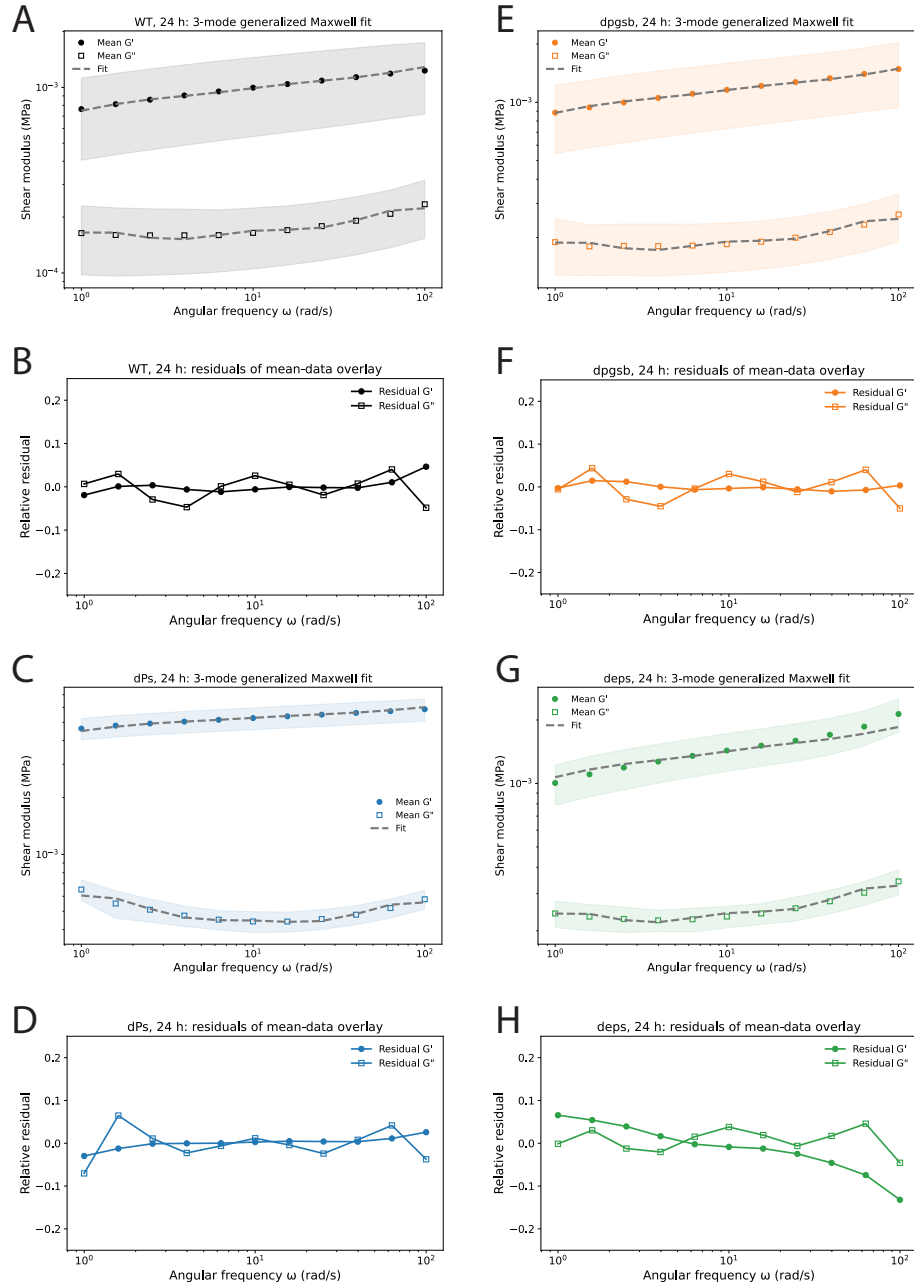
215 Figures S5 and S6 show the mean, standard deviations, fits, and rel-  
 216 ative residuals of the three-mode generalized Maxwell model with our  
 217 frequency sweep data. The fits and relative residuals are generally of  
 218 good quality. Each replicate was fit separately. The fitting parameters  
 219 were averaged and from the averages the fits were plotted.

#### 220 E. Strain sweep data

221 Figure S7 shows the mean storage and loss moduli and loss factor vs  
 222 strain amplitude with the shaded regions showing the standard devi-  
 223 ations. Plateau moduli were extracted from these data, the analysis of  
 224 which is described in the main text.

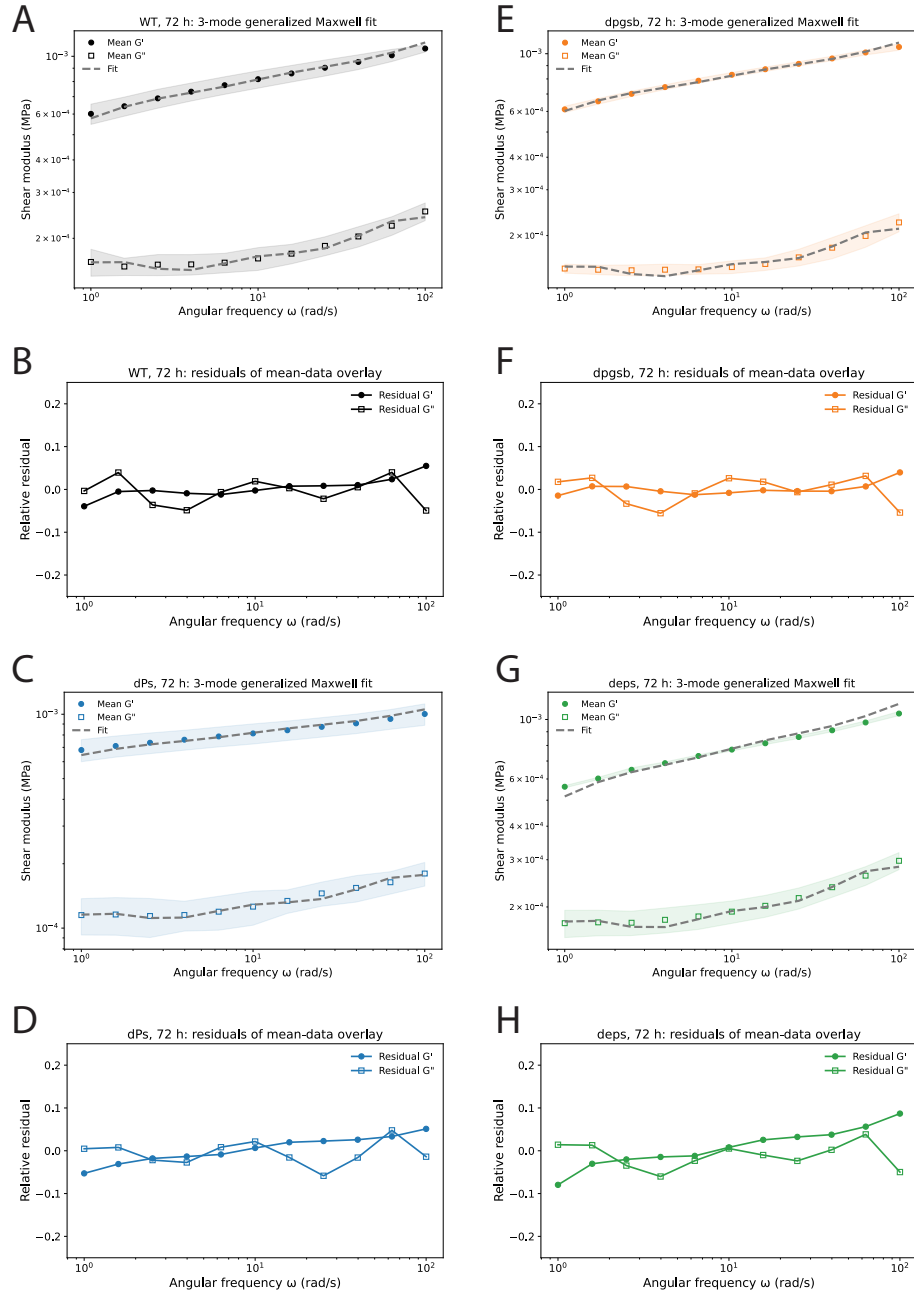
#### 225 F. Stereo microscope videos of biofilm scraping

226 We recorded videos using a stereo microscope as described in the main  
 227 text. These videos are accessible with the other supplemental materials



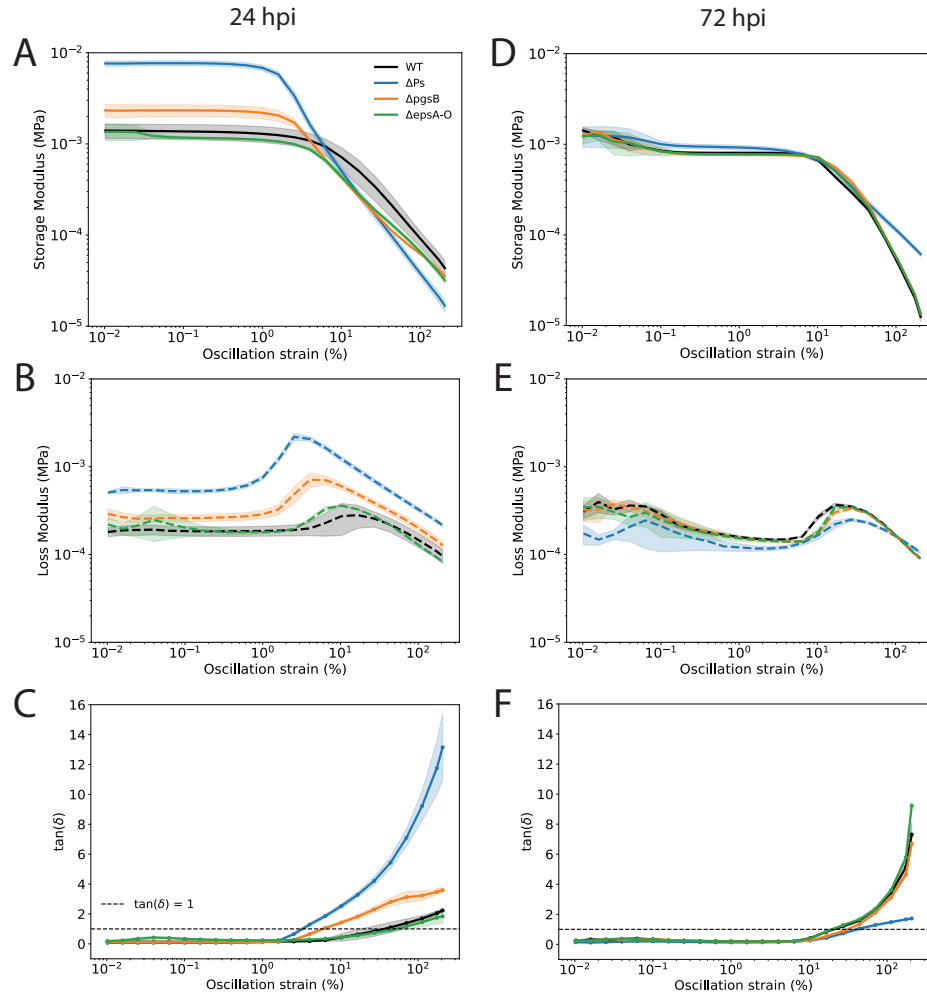
**Fig. S5.** Mean storage and loss moduli with fit to three-mode generalized Maxwell model at 24 hours for (A) WT, (C)  $\Delta P_s$ , (E)  $\Delta pgsB$ , and (G)  $\Delta epsA-O$  biofilms. Relative fit residuals (residual divided by modulus) for (A) WT, (C)  $\Delta P_s$ , (E)  $\Delta pgsB$ , and (G)  $\Delta epsA-O$  biofilms. Shaded regions indicate standard deviations. N = 3 independent samples per strain.

228 at <http://www.biophysj.org> and at [https://github.com/Larkin-Lab/ducharme\\_larkin\\_sporulation.git](https://github.com/Larkin-Lab/ducharme_larkin_sporulation.git).  
 229



**Fig. S6.** Mean storage and loss moduli versus angular frequency with fit to three-mode generalized Maxwell model at 72 hours for (A) WT, (C)  $\Delta P_s$ , (E)  $\Delta pgsB$ , and (G)  $\Delta epsA$ -Obiofilms. Relative fit residuals (residual divided by modulus) for (A) WT, (C)  $\Delta P_s$ , (E)  $\Delta pgsB$ , and (G)  $\Delta epsA$ -O biofilms. Shaded regions indicate standard deviations. N = 3 independent samples per strain.

230 The videos were recorded with the same exposure settings so any  
 231 differences in apparent brightness are due to physical differences between



**Fig. S7.** Mean storage moduli vs strain amplitude at (A) 24 and (D) 72 hours. Mean loss moduli at (B) 24 and (E) 72 hours. Mean loss factor at (C) 24 and (F) 72 hours. Shaded regions indicate standard deviations. N = 3 independent samples per strain.

232 biofilms.

233 We dragged the bottom of a micro-centrifuge tube through the center  
 234 of the biofilms, displacing a small volume of biomass. We then recorded  
 235 videos over the following three minutes. We observed a difference in the  
 236 recoil of the biofilms based on their ability to sporulate. The wild-type  
 237 (spore-producing) biofilm exhibited greater recoil, i.e., greater return  
 238 towards its pre-injured shape, compared to the spore-knockout biofilms.  
 239 This observation hinted at a difference in the rheological properties of the  
 240 biofilms based on sporulation ability, which led to the research presented  
 241 here.

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